

Class Activity

Date:24-07-26

1.Stacked Bars

Code:

```
# Employee Salary Data
department <- c("HR", "Finance", "IT", "Sales")
Male <- c(5, 6, 8, 7)
Female <- c(4, 5, 6, 5)
# Create matrix
emp_data <- rbind(Male, Female)
# Draw Stacked Bar Chart
barplot(emp_data,
        main = "Employees by Department",
        xlab = "Department",
        ylab = "Number of Employees",
        col = c("skyblue", "pink"),
        names.arg = department)
# Add Legend
legend("topright",
      legend = c("Male", "Female"),
      fill = c("skyblue", "pink"))
```

Output:



2.Pie Chart

Code:

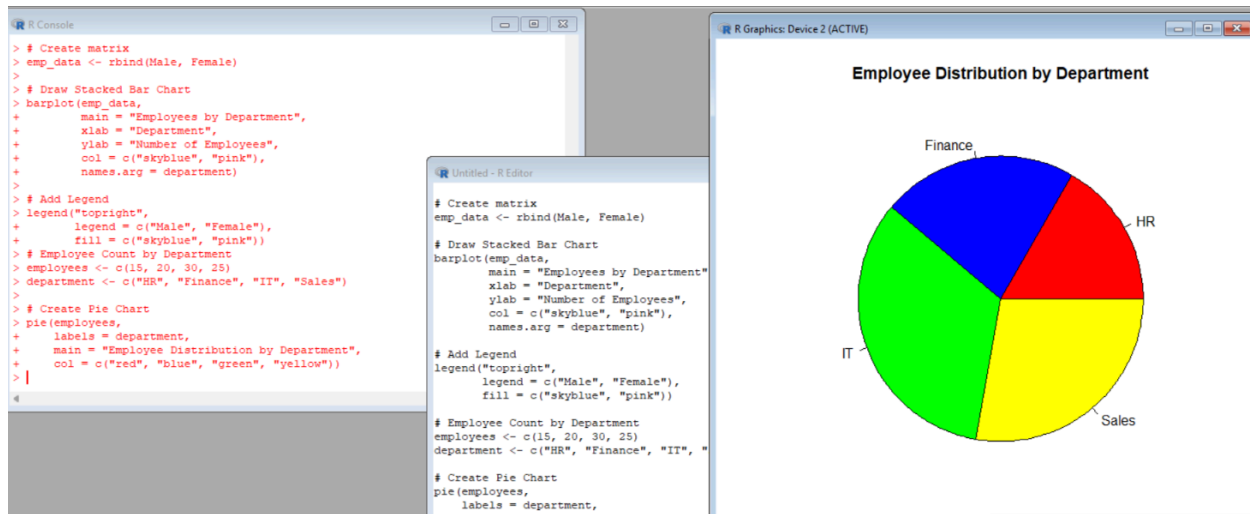
```
# Employee Count by Department
```

```

employees <- c(15, 20, 30, 25)
department <- c("HR", "Finance", "IT", "Sales")
# Create Pie Chart
pie(employees,
    labels = department,
    main = "Employee Distribution by Department",
    col = c("red", "blue", "green", "yellow"))

```

Output:



3. Treemaps

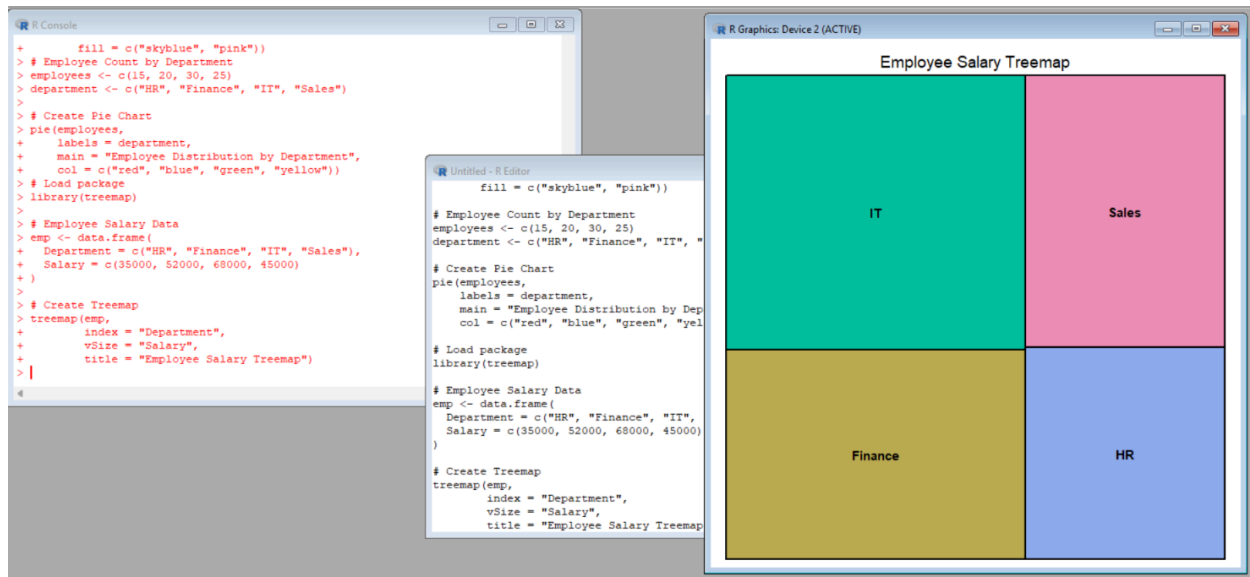
Code:

```

# Load package
library(treemap)
# Employee Salary Data
emp <- data.frame(
  Department = c("HR", "Finance", "IT", "Sales"),
  Salary = c(35000, 52000, 68000, 45000)
)
# Create Treemap
treemap(emp,
  index = "Department",
  vSize = "Salary",
  title = "Employee Salary Treemap")

```

Output:



4.X-Y Relationships

Code:

X and Y values

```
x <- c(10, 20, 30, 40, 50)
```

```
y <- c(15, 25, 35, 45, 55)
```

Scatter Plot

```
plot(x, y,
```

```
  main = "X-Y Relationship",
```

```
  xlab = "X Values",
```

```
  ylab = "Y Values",
```

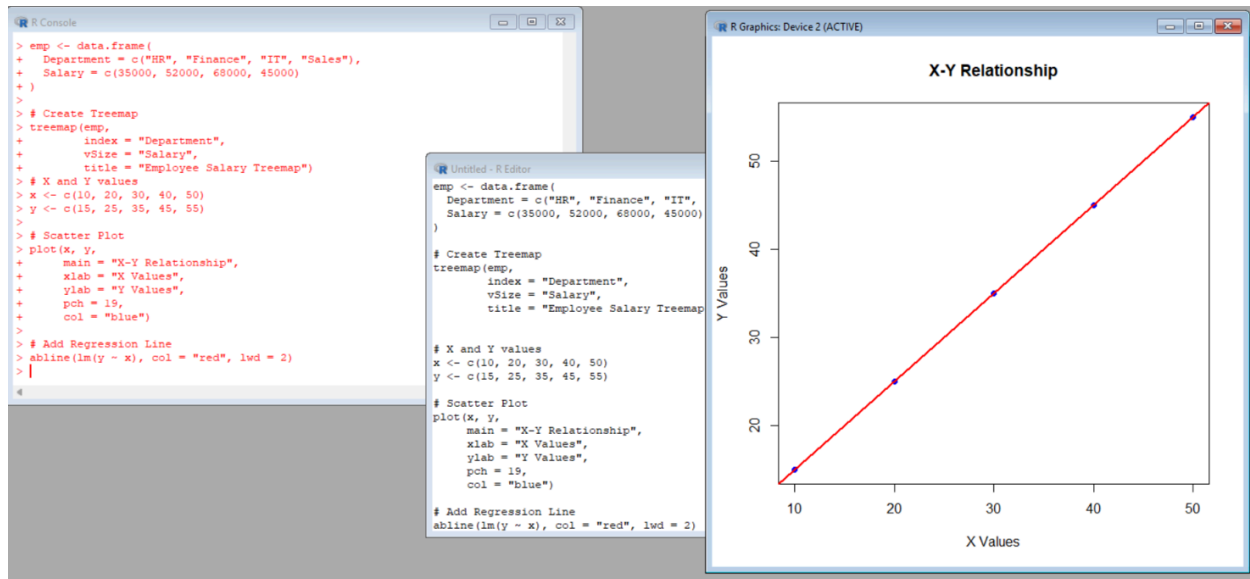
```
  pch = 19,
```

```
  col = "blue")
```

Add Regression Line

```
abline(lm(y ~ x), col = "red", lwd = 2)
```

Output:



5.Scatterplot

Code:

X and Y values

```
x <- c(10, 20, 30, 40, 50)
```

```
y <- c(15, 22, 35, 42, 55)
```

Scatter Plot

```
plot(x, y,
```

```
  main = "Scatter Plot",
```

```
  xlab = "X Values",
```

```
  ylab = "Y Values",
```

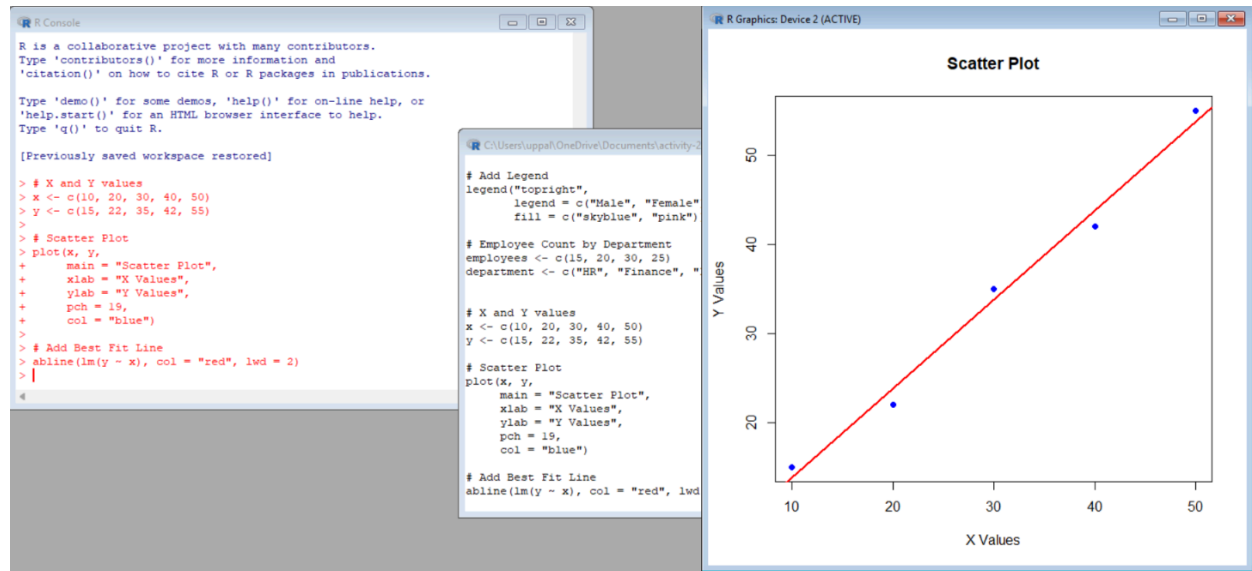
```
  pch = 19,
```

```
  col = "blue")
```

Add Best Fit Line

```
abline(lm(y ~ x), col = "red", lwd = 2)
```

Output:

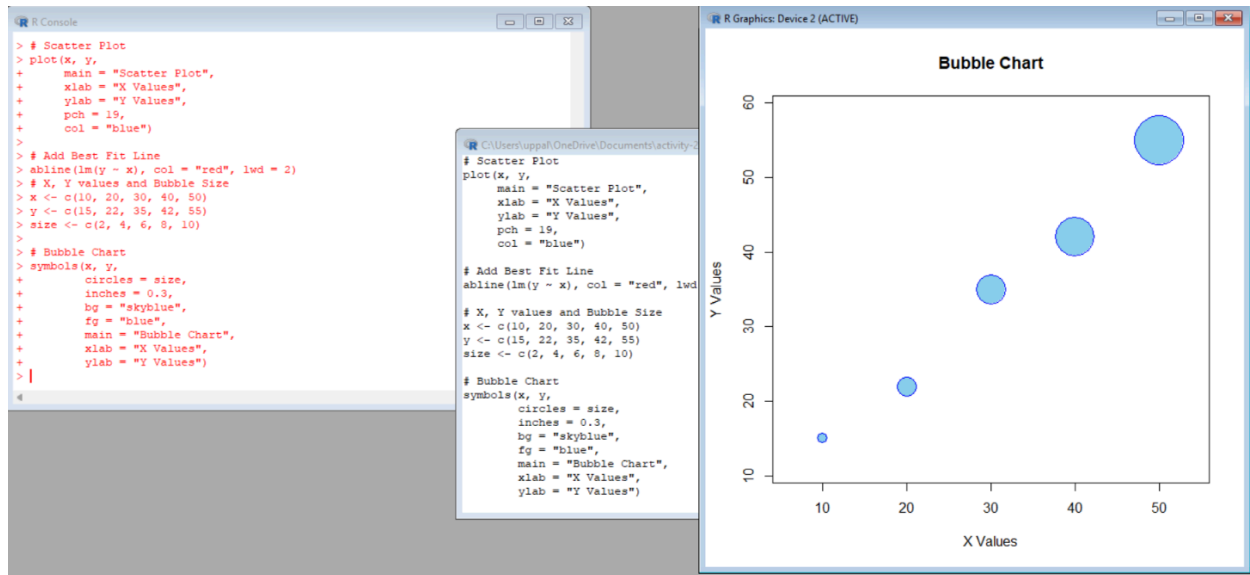


6. Bubble Chart

Code:

```
# X, Y values and Bubble Size
x <- c(10, 20, 30, 40, 50)
y <- c(15, 22, 35, 42, 55)
size <- c(2, 4, 6, 8, 10)
# Bubble Chart
symbols(x, y,
        circles = size,
        inches = 0.3,
        bg = "skyblue",
        fg = "blue",
        main = "Bubble Chart",
        xlab = "X Values",
        ylab = "Y Values")
```

Output:

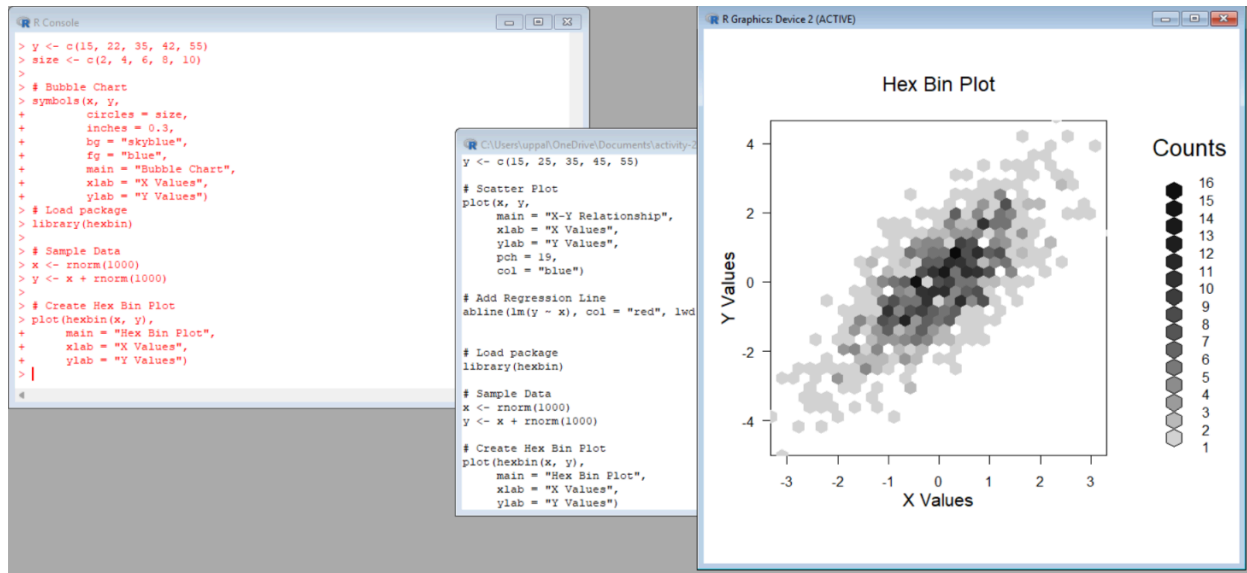


7.Hex Bin

Code:

```
# Load package
library(hexbin)
# Sample Data
x <- rnorm(1000)
y <- x + rnorm(1000)
# Create Hex Bin Plot
plot(hexbin(x, y),
     main = "Hex Bin Plot",
     xlab = "X Values",
     ylab = "Y Values")
```

Output:

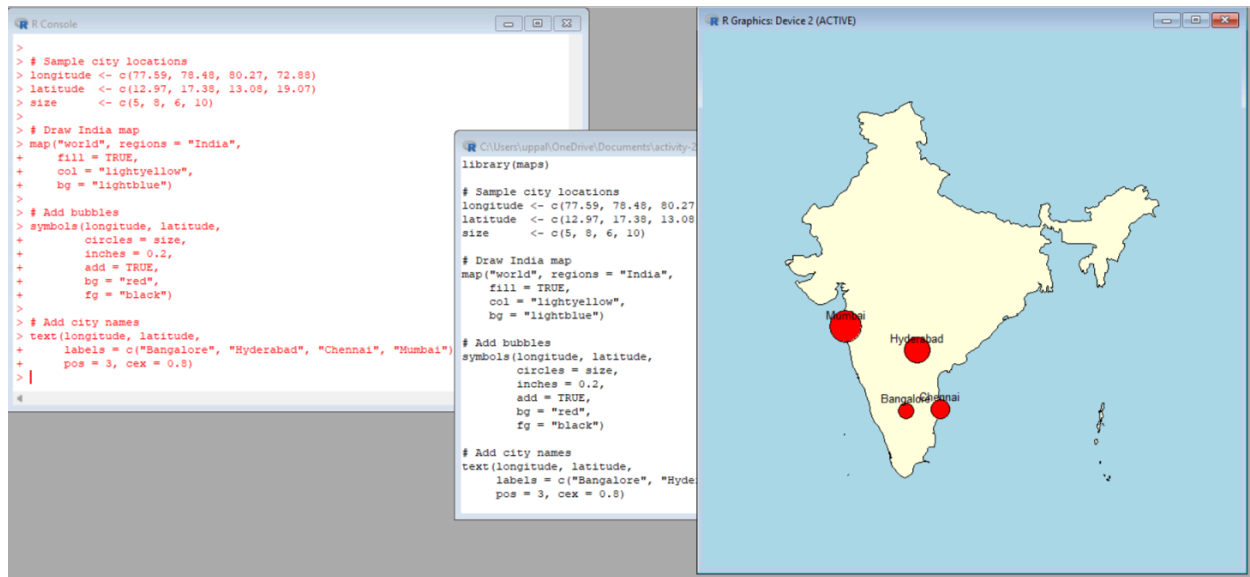


8. Bubble Map

Code:

```
library(maps)
# Sample city locations
longitude <- c(77.59, 78.48, 80.27, 72.88)
latitude <- c(12.97, 17.38, 13.08, 19.07)
size <- c(5, 8, 6, 10)
# Draw India map
map("world", regions = "India",
    fill = TRUE,
    col = "lightyellow",
    bg = "lightblue")
# Add bubbles
symbols(longitude, latitude,
        circles = size,
        inches = 0.2,
        add = TRUE,
        bg = "red",
        fg = "black")
# Add city names
text(longitude, latitude,
     labels = c("Bangalore", "Hyderabad", "Chennai", "Mumbai"),
     pos = 3, cex = 0.8)
```

Output:



9.Connected Scatterplots

Code:

X and Y values

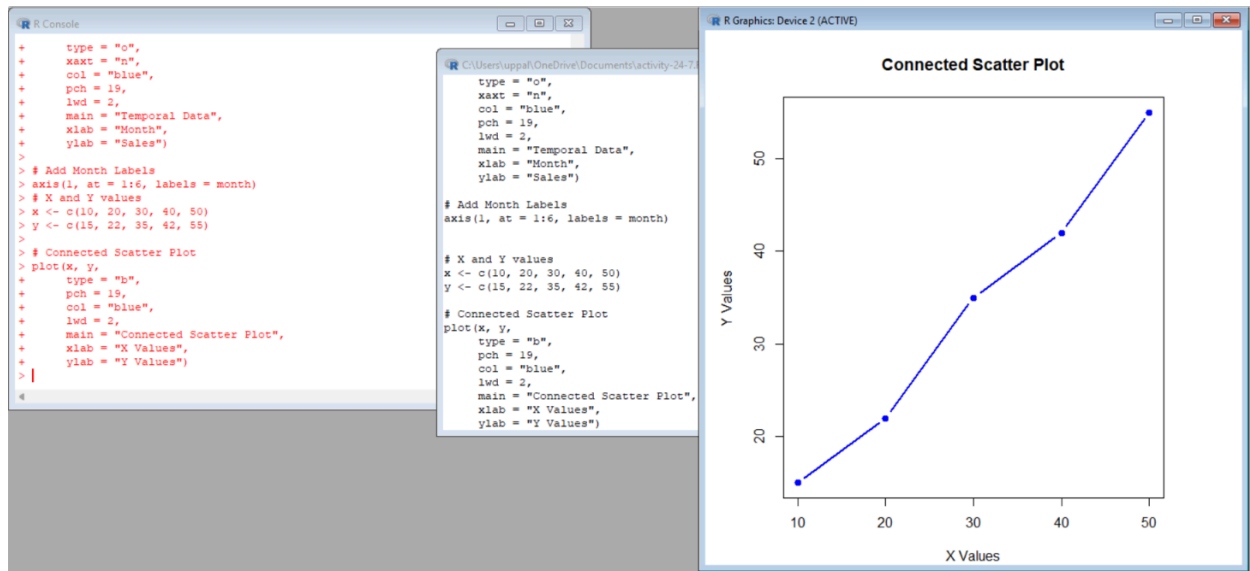
```
x <- c(10, 20, 30, 40, 50)
```

```
y <- c(15, 22, 35, 42, 55)
```

Connected Scatter Plot

```
plot(x, y,
     type = "b",
     pch = 19,
     col = "blue",
     lwd = 2,
     main = "Connected Scatter Plot",
     xlab = "X Values",
     ylab = "Y Values")
```

Output:



10. Network Graph

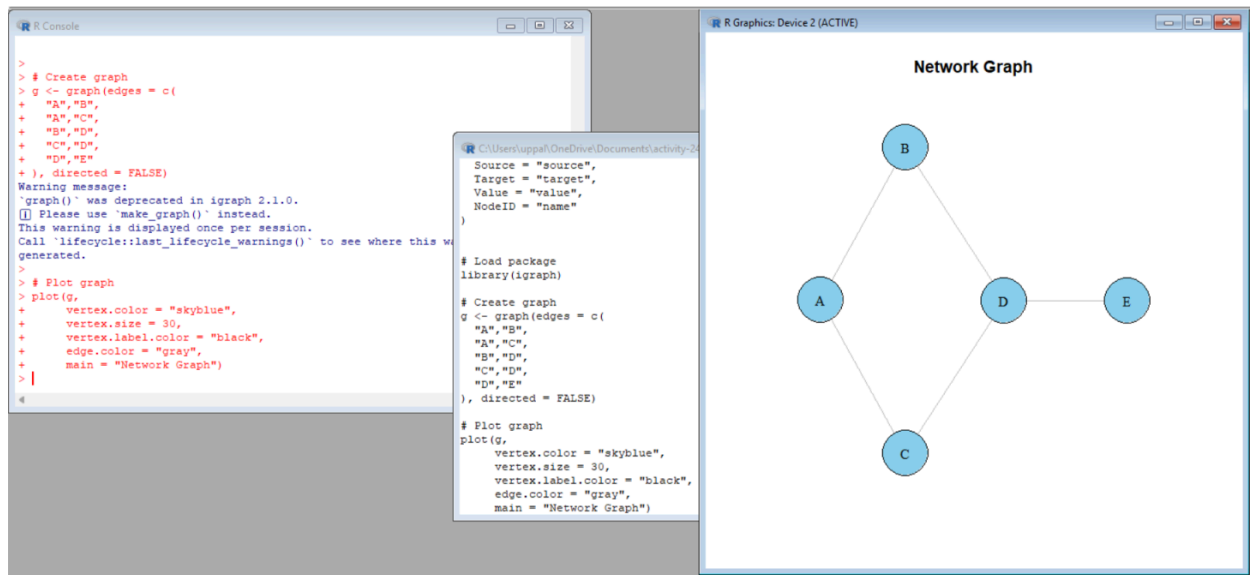
Code:

```

# Load package
library(igraph)
# Create graph
g <- graph(edges = c(
  "A", "B",
  "A", "C",
  "B", "D",
  "C", "D",
  "D", "E"
), directed = FALSE)
# Plot graph
plot(g,
  vertex.color = "skyblue",
  vertex.size = 30,
  vertex.label.color = "black",
  edge.color = "gray",
  main = "Network Graph")

```

Output:



11.Line Graph

Code:

X and Y values

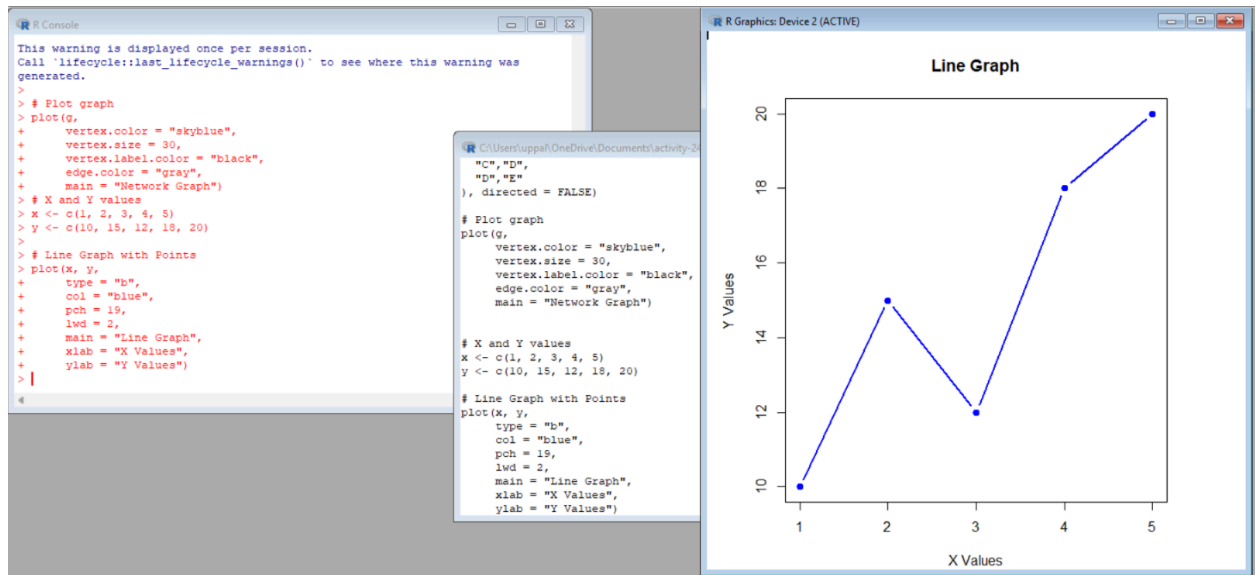
```
x <- c(1, 2, 3, 4, 5)
```

```
y <- c(10, 15, 12, 18, 20)
```

Line Graph with Points

```
plot(x, y,
     type = "b",
     col = "blue",
     pch = 19,
     lwd = 2,
     main = "Line Graph",
     xlab = "X Values",
     ylab = "Y Values")
```

Output:



12.Stream Graph

Code:

Sample Data

```
marks <- c(45, 48, 52, 56, 58, 61, 64, 67, 72, 75, 78, 82, 85, 88, 91)
```

Stem-and-Leaf Plot

```
stem(marks)
```

Output:

```

> # Sample Data
> marks <- c(45, 48, 52, 56, 58, 61, 64, 67, 72, 75, 78, 82,
+
> # Stem-and-Leaf Plot
> stem(marks)

The decimal point is 1 digit(s) to the right of the |

 4 | 58
 5 | 268
 6 | 147
 7 | 258
 8 | 258
 9 | 1
>

```

13.Temporal Data

Code:

Time and Sales data

```

month <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun")
sales <- c(120, 150, 170, 160, 190, 210)
# Line Graph
plot(sales,
     type = "o",
     xaxt = "n",
     col = "blue",
     pch = 19,
     lwd = 2,
     main = "Temporal Data",
     xlab = "Month",
     ylab = "Sales")
# Add Month Labels
axis(1, at = 1:6, labels = month)

```

Output:

